Predictive Analytics in Public Transit: A National Framework for Safety Risk Mitigation

Author: Chinonso Eziefule

Subject: Open-Source Safety Management System (SMS) Architecture for Federal Transit Administration (FTA) Compliance

Executive Summary

The United States public transportation sector is currently navigating a critical inflection point regarding the safety and security of its frontline workforce. In direct response to the escalating rates of severe safety events, the Federal Transit Administration (FTA) issued General Directive 24-1, mandating rigorous, data-driven Safety Risk Assessments across all applicable transit agencies. This paper outlines the architecture and deployment of an open-source, AI-driven Safety Management System (SMS) designed to transition the industry from reactive incident reporting to proactive, predictive risk mitigation. Utilizing XGBoost and SHAP explainable AI, this framework transforms National Transit Database (NTD) records into actionable intelligence, providing a scalable blueprint for transit authorities nationwide.

1. The Federal Mandate and the Industry Gap

Historically, regional transit authorities have relied on legacy, manual methodologies; specifically the MIL-STD-882E 5x5 Risk Matrix to evaluate hazard probabilities and severities. While effective for localized, static environments, these spreadsheet-based tools are fundamentally unsuited for analyzing the high-volume, dynamic data sets generated by modern urban transit networks.

When the FTA mandated comprehensive safety risk assessments, mid-to-large-sized agencies faced a significant technological gap. Many lack the internal data engineering infrastructure required to continuously ingest, cleanse, and analyze massive federal datasets. Consequently, the industry requires a modernized, standardized framework that can accurately forecast risk without imposing unsustainable engineering burdens on local municipalities.

2. Architectural Methodology: AI-Driven Risk Prediction

To solve this infrastructure gap, a complete, cloud-native predictive data pipeline was engineered. The architecture operates through a continuous, automated cycle:

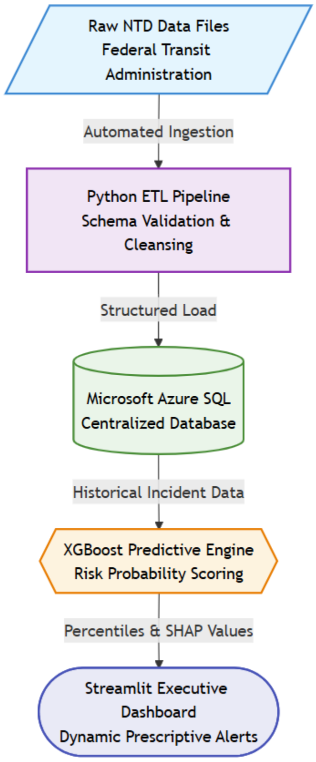

Figure 1: The System Architecture Pipeline

a) **Data Engineering (ETL):** An automated Python pipeline extracts raw safety, security, and geographic data directly from the public NTD portal. The pipeline enforces strict schema validation and duplicate-cleansing protocols before loading the data into a centralized Microsoft Azure SQL Database, ensuring that dirty federal data does not artificially inflate regional risk profiles.

b) **Machine Learning Engine**: The system deploys an XGBoost Regressor, trained extensively on historical incident severity data (including fatalities, injuries, and assaults). By analyzing temporal, environmental, and operational features, the algorithm calculates forward-looking risk probabilities, establishing a dynamic national baseline for transit safety.

c) **Executive Dashboarding:** The predictive outputs are routed to a secure, interactive Streamlit web application, allowing regional directors to visualize the data through geospatial heatmaps and longitudinal trend analyses.

3. Explainable AI and Executive Translation

Executive Summary: This agency is classified as a **Tier 1 Critical Risk** (96.6th Percentile nationally). The primary driver is **Recent Assault Trends**, adding **+2.1 points** above baseline.

Figure 2: SHAP Waterfall Analysis of Agency-Specific Risk Drivers

A persistent barrier to AI adoption in federal and municipal government is the "black box" phenomenon, where algorithms output directives without providing the mathematical reasoning. To achieve federal compliance and ensure executive trust, this system utilizes SHAP (SHapley Additive exPlanations).

The framework dynamically generates waterfall visualizations that break down the exact operational drivers influencing an agency's risk score. Instead of merely outputting a raw risk percentage, the system explains the variance from the national baseline; isolating variables such as recent assault trends, worker fatigue indicators, or environmental hazards.

Furthermore, to align with existing federal safety culture, the system translates these advanced machine learning percentiles into dynamic command recommendations. Agencies falling within the top 25% (Elevated Risk) or top 5% (Critical Risk) nationally receive automated, prescriptive resource allocation strategies, such as dynamic patrol redeployments or operator schedule audits.

4. National Impact and Adoptability

The primary objective of this framework is widespread, frictionless adoption. The architecture was specifically designed as a "plug-and-play" repository. By utilizing standard containerization requirements, any data analyst at a regional transit authority can clone the repository, connect it to their localized incident database, and train the predictive engine on their own historical data.

Conclusion

Protecting transit workers requires more than static policy; it requires modern, scalable data infrastructure. This open-source predictive framework directly satisfies the mandates of FTA General Directive 24-1 by providing a mathematically rigorous, fully transparent, and highly adoptable Safety Management System. By lowering the barrier to entry for advanced predictive analytics, this architecture establishes a new, proactive standard for mitigating transit worker risk across the United States.